AI WEATHER WISE

AI-POWERED WEATHER FORECASTING, INSIGHTS & CLIMATE ANALYTICS API

PROJECT TEAM

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PROJECT DESCRIPTION

AI WEATHER WISE is a robust, secure, and intelligent REST API platform engineered for real-time weather analytics, AI-assisted forecasting insights, user preference management, and automated climate reporting. Built using **Node.js** and **Express.js** with **MongoDB (Mongoose ODM)**, it integrates external live weather APIs with **Google Gemini AI** for personalized recommendations and extreme hazard alerts.

SCENARIO-BASED CASE STUDY

Background & Problem

Arun, an outdoor event organizer, struggles with raw meteorological data to plan safe activities and agricultural operations.

- Difficulty translating raw data into clear actions.
- Limited visibility into local severe weather shifts.
- Time-consuming manual report drafting.
- Lack of secure permission layers for API consumers.

Solution & Outcome

A centralized API gateway decoupling raw ingestion from client consumption with built-in AI intelligence.

- **Automated AI Advisories:** Instant hazard scoring.
- **Decoupled Architecture:** Offloads heavy processing.
- **Role-Based Security:** Strict JWT access control.
- **Fast Retrieval:** Indexed MongoDB collections.

SYSTEM REQUIREMENTS

Software Stack

Node.js v18+ | Express.js v4/v5 | MongoDB / Mongoose | JWT & bcrypt | Google Gemini AI | Postman / ThunderClient

Hardware Infrastructure

- **Processor:** Intel Core i5 (8th Gen+) / AMD Ryzen 5
- **Memory:** 8 GB RAM (16 GB recommended)
- **Storage:** 1 GB available disk space

EPIC 1: TECHNICAL ARCHITECTURE FLOW

```
    +-------------------------------------------------------------------------------+ | CLIENTS |
```

ENTITY-RELATIONSHIP (ER) ARCHITECTURE

```
    +-----------------+ 1 : N +---------------------+ | USER | ------------------> | WEATHER_REPORT |
```

KEY SYSTEM ROLES & CAPABILITIES

- **Admin:** Full access to manage users, system logs, public advisories, and global alert settings.
- **Content Creator / Meteorologist:** Issues weather advisories, triggers AI insight generation pipelines, and manages regional logs.
- **Authenticated Consumer:** Fetches live forecasts, requests personalized AI activity safety ratings, and manages saved locations.
- **Public User:** Accesses read-only public endpoints for current weather conditions.

DATA SCHEMAS & CORE CODE

1. User Model (`src/models/User.js`)

```
const mongoose = require('mongoose'); const bcrypt = require('bcrypt'); const userSchema = new mongoose.Schema({ name: { type: String, required: true, trim: true }, email: { type: String, required: true, unique: true, lowercase: true }, password: { type: String, required: true, minlength: 6 }, role: { type: String, enum: ['admin', 'creator', 'user'], default: 'user' } }, { timestamps: true }); userSchema.pre('save', async function (next) { if (!this.isModified('password')) return next(); this.password = await bcrypt.hash(this.password, 10); }); module.exports = mongoose.model('User', userSchema);
```

2. Auth Middleware (`src/middleware/authMiddleware.js`)

```
const jwt = require("jsonwebtoken"); const User = require("../models/User"); const protect = async (req, res, next) => { let token; if (req.headers.authorization?.startsWith("Bearer")) { try { token = req.headers.authorization.split(" ")[1]; const decoded = jwt.verify(token, process.env.JWT_SECRET); req.user = await User.findById(decoded.id).select("-password"); return next(); } catch (error) { return res.status(401).json({ success: false, message: "Token failed" }); } } return res.status(401).json({ success: false, message: "No token provided" }); }; module.exports = { protect };
```

API EXECUTION SPECIFICATIONS

Register User

`POST /api/auth/register`

```
{ "name": "Deepika K", "email": "deepika@weatherwise.com", "password": "securepassword123" }
```

Generate AI Insight

`POST /api/ai/generate-insight`

```
{ "location": "Chennai", "activity": "Outdoor Marathon", "forecastData": "33°C, 80% humidity" }
```